Lab 08 - Cluster and classify reviews

Course: TGS-2026064715 | Maps to: K5 K6 K7; A5 A6

Goal

Compare a K-means elbow trace with a supervised classification baseline.

Data

hotel_reviews.csv contains 120 fictional Singapore hotel and service reviews with dates, channel, rating, sentiment and property. Names and review text are synthetic; no customer records or external credentials are used.

Steps

1. Open a terminal in this lab folder. Confirm Python 3.10+ with python3 --version.
2. Install the shared requirements from the parent labs/requirements.txt with python3 -m pip install -r ../requirements.txt.
3. Inspect the first rows and column names in hotel_reviews.csv. Check that review_id is unique and sentiment has three classes.
4. Run python3 run.py from this folder. Read the printed output and any files created in output/.
5. Explain how the result supports the lab goal, what a production dataset would add, and one failure or bias risk.

Acceptance check

The script exits with code 0. Verify that the reported row count is 120. For model labs, inspect the holdout scores rather than training accuracy; for summaries, inspect the exported CSV/PNG.

Reflection

How would different hotel review wording, label quality, or data leakage change this result?